UML AASX Studio

Class Studio — UML Class Diagram

User Manual

Semi-automatic generation of AAS Submodel templates from UML class diagrams

Document version: 1.2

Based on the UML2AAS tool

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1. Introduction

UML AASX Studio is a desktop application that implements the UML2AAS tool, developed to semi-automatically generate Asset Administration Shell (AAS) Submodel templates from UML class diagrams. The tool translates the classes, attributes, associations, generalizations and enumerations of a UML model into the corresponding elements of the AAS metamodel, applying a set of formal transformation rules.

It is intended for domain specialists and engineers who need to produce reusable, standard-compliant AAS Submodels starting from a familiar conceptual model (UML) instead of building the AASX file directly.

1.1 Class Studio module

The main working module is called Class Studio. In it, the UML class diagram is drawn on a canvas, the model is validated, and the result is exported in different formats, including AASX.

1.2 Supported formats

- **Input:** The tool's native JSON, Draw.io (.drawio or PNG with diagrams.net data) and XML/XMI (XMI 2.1 and 2.5 are detected automatically).
- **Output:** Native JSON, XMI (version 2.1 or 2.5), AASX (AAS version 3.0.1 or 3.1.2) and an image of the full diagram (SVG or JPEG).

1.3 What's new in version 1.2 of the manual

This version of the manual covers the features recently added to Class Studio:

- **Arrange:** automatic layered reorganization of the diagram (section 3.3).
- **Export image:** export of the full diagram to SVG or JPEG (section 5.4).
- **Custom multiplicity:** minimum and maximum editor for ranges such as 2..5 or 2..* (section 4.5).
- **Manual routes:** relations adjusted by hand or imported keep their shape and can be returned to automatic routing (section 4.8).
- **Redesigned validation panel:** collapsible cards, one-step Apply all and an issue navigator in collapsed mode (section 6).
- **Report & Feedback:** form for reporting problems or suggesting improvements (section 9).

2. Getting started

When you open UML AASX Studio and enter the Class Studio module, a start screen is shown with three options to begin working.

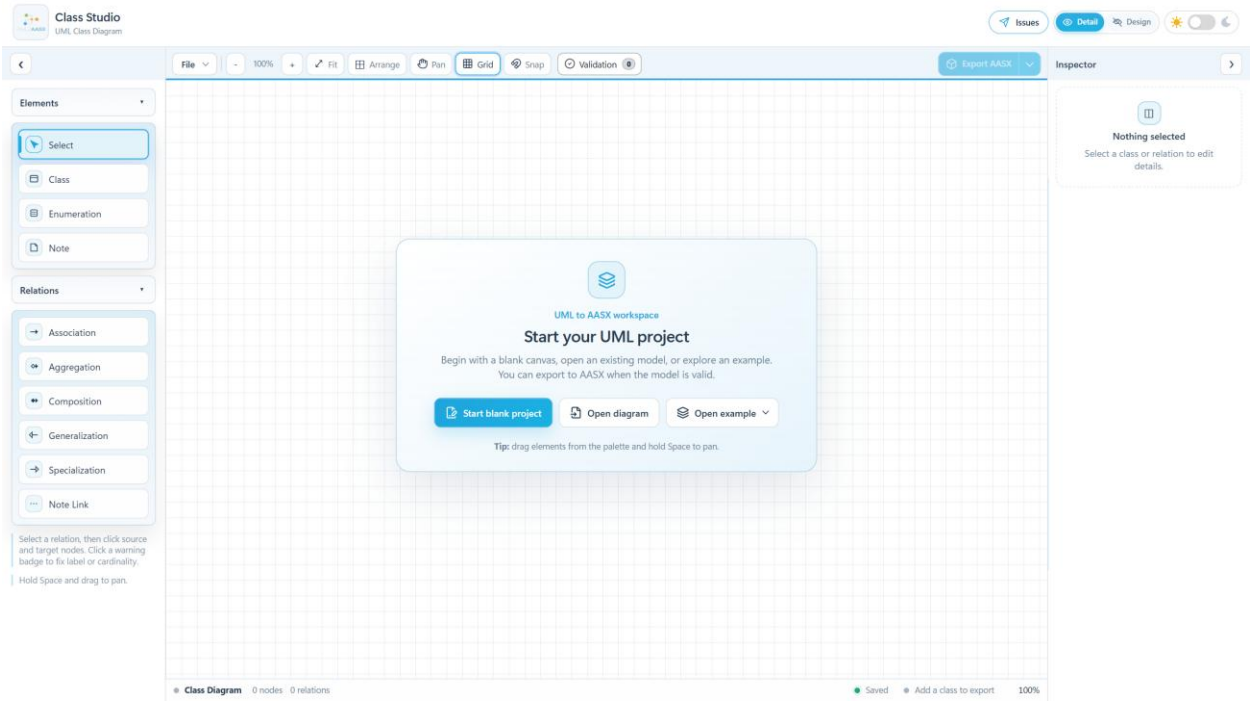


Figure 1. Class Studio start screen.

- **Start blank project:** opens an empty canvas to model from scratch.
- **Open diagram:** imports an existing diagram in JSON, Draw.io or XML/XMI.
- **Open example:** expands the list of example diagrams included with the tool. Each example shows a short description of what it illustrates.

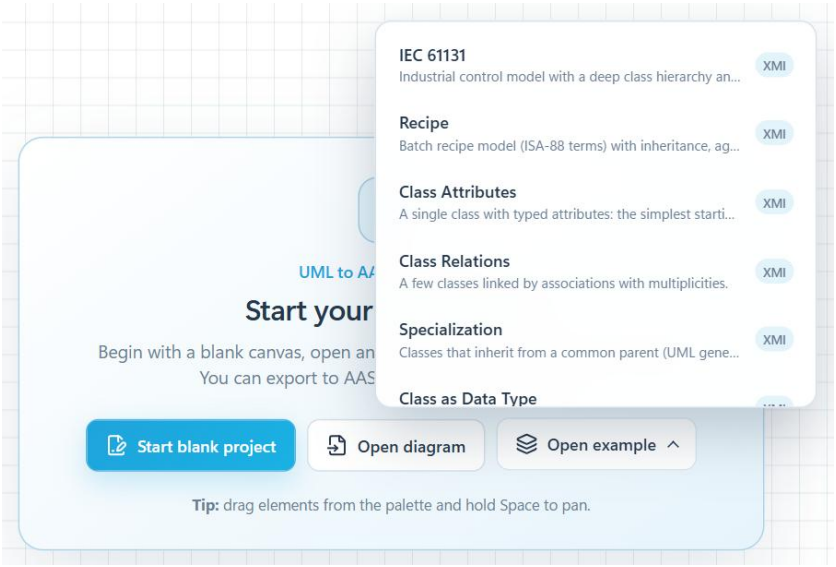


Figure 2. List of examples available from the start screen.

The available examples are:

- **IEC 61131:** industrial control model with a deep class hierarchy and composition.
- **Recipe:** batch recipe model (ISA-88 terminology) with inheritance, aggregations and enumerations.
- **Class Attributes:** a single class with typed attributes; this is the simplest starting point.

- **Class Relations:** several classes linked by associations with multiplicities.
- **Specialization:** classes that inherit from a common parent class (UML generalization).
- **Class as Data Type:** a class reused as the data type of an attribute of another class.

> **Note:** hold down the Space key and drag to pan across the canvas. Elements are added by dragging them from the side panel.

The start screen is only shown when the application is opened. If you later create a new project or delete all elements, the empty canvas is ready for editing without showing it again.

3. General interface

The Class Studio window is organized into five areas: the application's top bar, the left side panel (Elements / Relations), the canvas toolbar, the central canvas with its status bar, and the Inspector panel on the right.

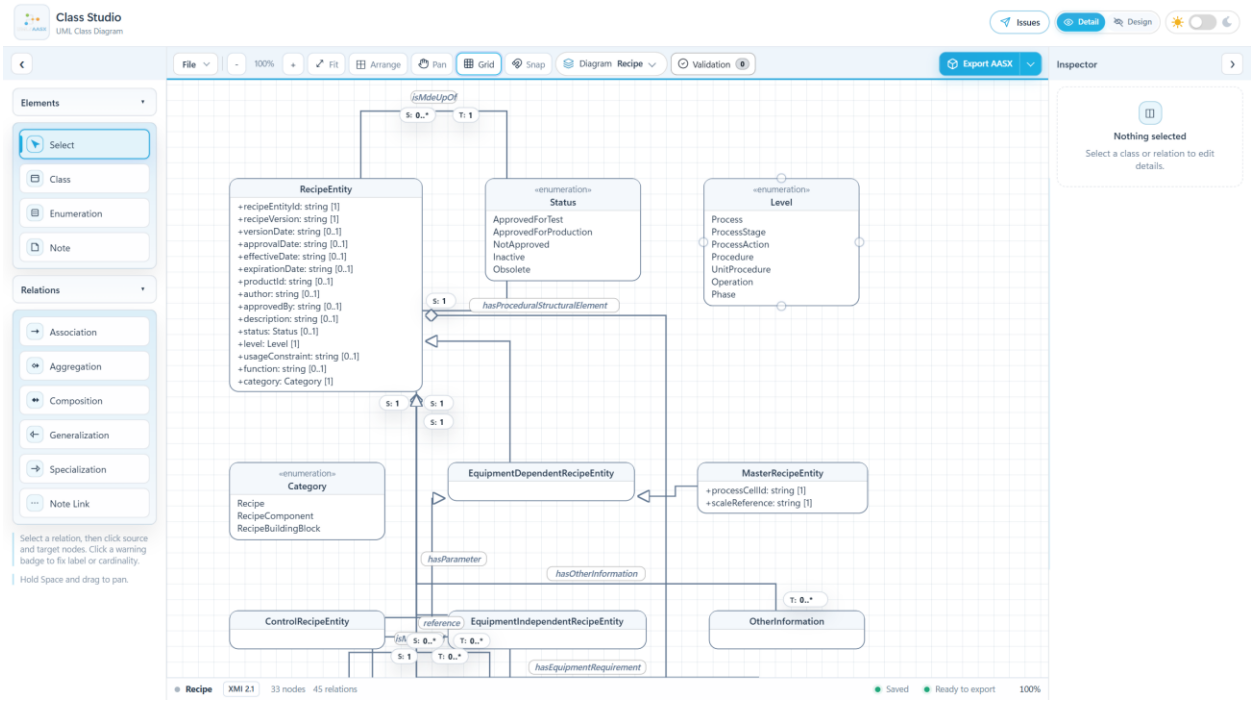


Figure 3. Overview of Class Studio with the Recipe example open.



Figure 4. Side panel with the Elements and Relations sections.

3.1 Elements panel

It contains the basic building blocks that can be dragged onto the canvas:

- **Select:** selection tool (default).
- **Class:** creates a UML class.
- **Enumeration:** creates an enumeration.
- **Note:** adds a note or text comment to the diagram.

3.2 Relations panel

It includes the relation types available between classes:

- Association
- Aggregation
- Composition
- Generalization
- Specialization
- **Note Link:** connects a note to the element it refers to.

> **Note:** to draw a relation, select the relation type in the panel, click the source node and then the target node. Escape cancels a connection in progress. Relation issues (label or multiplicity) are reviewed and fixed from the Validation panel (see section 6).

3.3 Top toolbar

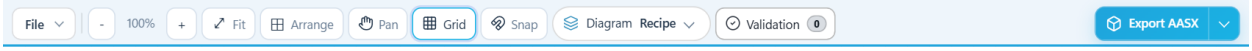


Figure 5. Canvas toolbar: File menu, zoom, Fit, Arrange, Pan, background, Snap, diagram selector, Validation and Export AASX.

From left to right, the top bar offers:

- **File:** workspace, import, save and export menu (see section 5).
- **Zoom controls (−, %, +) and Fit:** adjust the zoom level; Fit frames the whole diagram in the view.
- **Arrange:** automatically reorganizes the classes in layers: parent classes (and owners of compositions and aggregations) at the top, their children below, classes that only have associations next to their neighbours, and classes without relations in a final row. Manually adjusted routes return to automatic routing. The whole reorganization is undone in a single step with Ctrl+Z.
- **Pan:** enables canvas panning mode with the mouse.
- **Grid / Dots / Off: toggles the background between a line grid, dots and off.**
- **Snap: snaps the movement of nodes and relation segments to 20-pixel steps. Alignment guides are purely visual.**
- **Diagram:** shows the name of the active diagram. When the imported file contains several diagrams, the drop-down lets you switch between them.
- **Validation:** shows the total number of model issues (an ellipsis while the check is in progress) and opens the validation panel (see section 6). The button colour indicates whether there are blocking errors (red) or only warnings (amber). It is only shown in Detail mode.
- **Export AASX:** exports the model to AASX format with the AAS version shown on the button; the adjacent arrow lets you choose another version (see section 7). The button is disabled while the diagram has no elements and is shown in red when there are blocking errors; in that case, clicking it opens the validation panel instead of exporting.

The bottom status bar shows the diagram name, the source format when the model was imported (for example, XMI 2.1), the number of nodes and relations, the save status (Saved or Unsaved changes), the AASX status (for example, Ready to export or the number of blocking errors) and the zoom level. Clicking the AASX status opens the validation panel.

Figure 6. Status bar with the Recipe diagram imported from XMI 2.1.

3.4 Inspector panel

On the right of the screen, the Inspector shows the properties of the element selected on the canvas (class, attribute or relation). When nothing is selected, it displays "Nothing selected". It can be collapsed with the arrow in its top corner. Detailed use of the Inspector is described in section 4.

3.5 Detail / Design mode

In the top right corner, the Detail and Design buttons switch between the technical view of the model (the default, used for modelling and validating) and a design/presentation view of the diagram. The sun/moon switch next to them toggles between light and dark theme.

In Design mode the technical aids on the canvas (validation, quick actions, error indicators and multiplicities) are hidden, but the model is not modified. Go back to Detail to fix or validate the diagram.

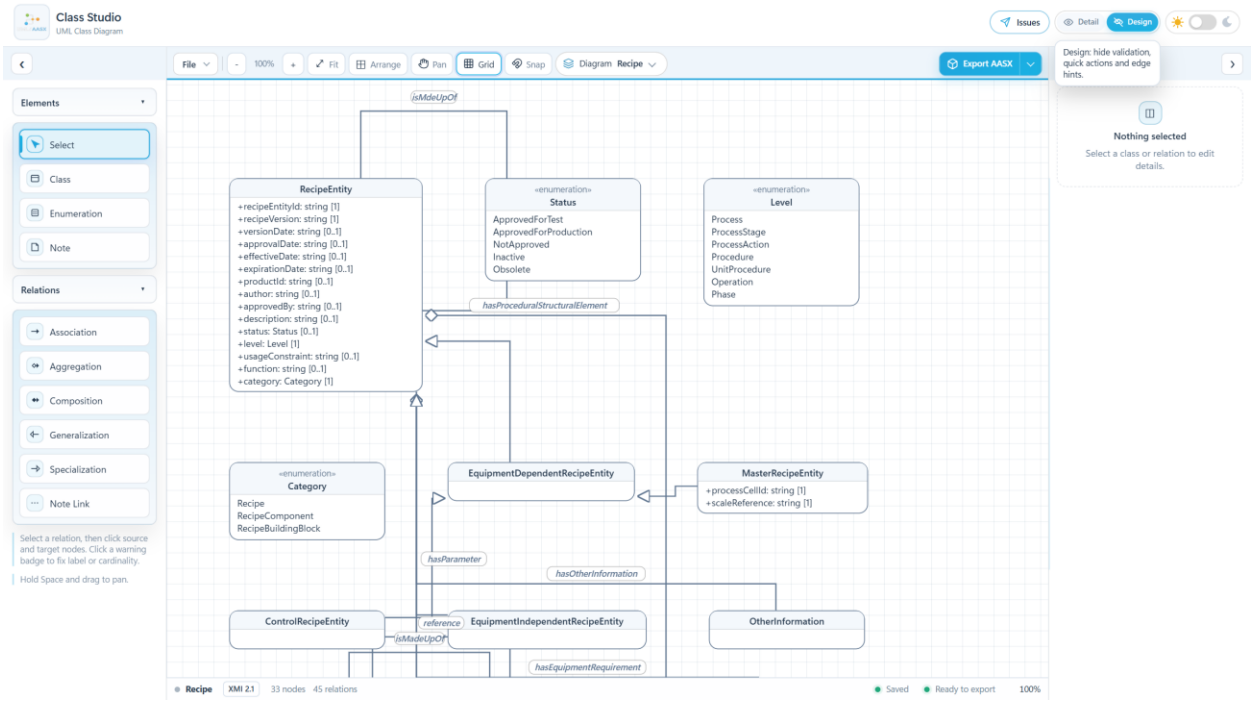


Figure 7. The Recipe example in Design mode: no multiplicities, quick actions or Validation button.

3.6 Issues button

The **Issues**button, in the top right corner, opens the **Report & Feedback** form to report a problem or suggest an improvement (see section 9). It is not related to the model validation issues, which are reviewed with the Validation button.

3.7 Collapsible panels and narrow windows

- The Elements/Relations and Inspector panels are collapsed and expanded with the arrows in their top corner, to leave more room for the canvas.

- When the canvas is narrow, the toolbar switches to showing only icons (with their name as a tooltip) and wraps onto several rows instead of hiding buttons.
- If the window is made very small (for example, half screen), both panels collapse automatically and open as drawers overlaid on the canvas. When the window is enlarged again, the previous state is restored.

4. Creating and editing the class diagram

4.1 Adding classes, enumerations and notes

1. To create an element at a default position, click Class, Enumeration or Note in the Elements panel.
2. To place it directly at a specific position, drag the element from the panel and drop it on the canvas.
3. The element is selected automatically and its properties appear in the Inspector, on the right.

4.2 Navigating, selecting and moving elements

- Click a node or a relation to select it and show its properties in the Inspector.
- Drag over an empty area to create a selection rectangle. Hold Shift to add nodes to the existing selection.
- Drag a node to move it. If it belongs to a multiple selection, the whole group moves.
- With Snap enabled, movement snaps to the grid. Guide lines help with visual alignment but do not attract the nodes.
- Use the handle in the bottom right corner of the selected node to resize it; notes do not have this handle.
- Drag while holding Space, enable Pan or use the middle button to pan the view. The mouse wheel zooms around the pointer.

4.3 Direct editing on the canvas

- Double-click the name of a class or enumeration to edit it without opening another dialog.
- Double-click an attribute to edit its UML expression. Confirm with Enter or by clicking outside; Escape cancels the edit.
- In a note, click the header to edit its name or the body to edit its text.

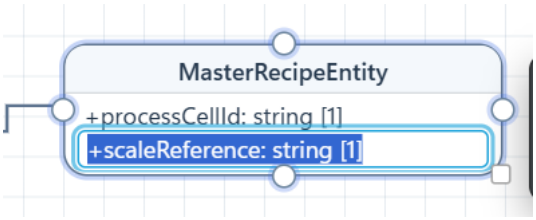


Figure 8. Direct editing of an attribute with a double-click.

- In Detail mode, selecting a class shows a quick-actions panel next to it: add an attribute (◧+), create a self-association (∞), a self-aggregation (◇) or a self-composition (◆).

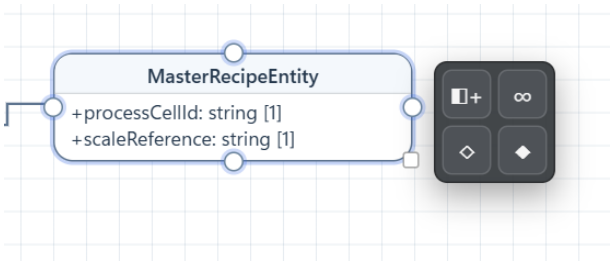


Figure 9. Selected class with the quick-actions panel.

To configure all fields and avoid syntax errors, use the Inspector as the main editing method.

4.4 Editing properties in the Inspector — General tab

When a class is selected on the canvas, the General tab of the Inspector lets you configure:

- **Name:** name of the class.
- **Stereotype:** UML stereotype associated with the class (drop-down menu).
- **Multi-instance:** switch that indicates whether the class should be exported as a collection of Submodel Elements (SMC List) instead of a single Submodel Element Collection (SMC).

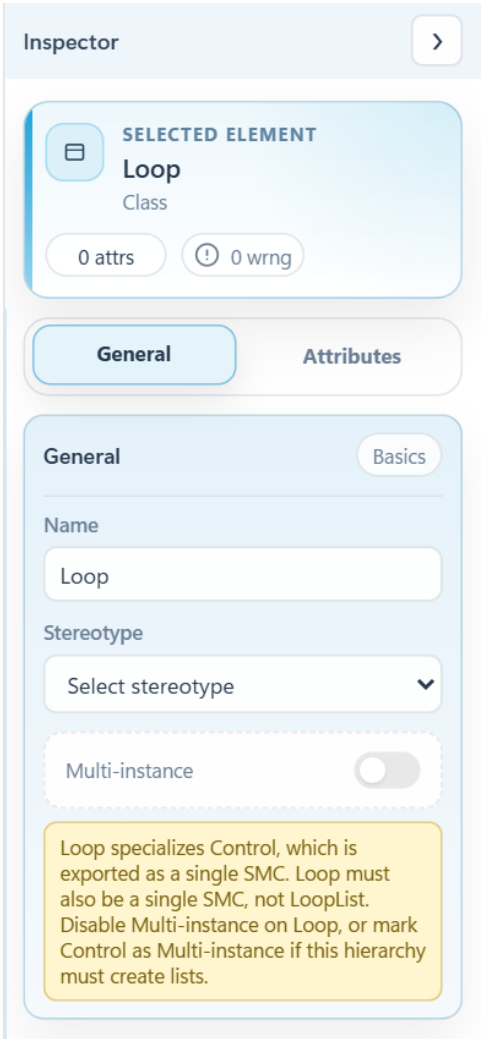


Figure 10. Inspector, General tab, with a hierarchy consistency warning about Multi-instance (Loop class of the IEC 61131 example).

> **Note:** the Inspector can show contextual warnings about the element's configuration. In the example in Figure 10, the Loop class specializes Control; since Control is exported as a single SMC (it is not marked as Multi-instance), Loop cannot be either, otherwise incompatible lists would be generated on export. That is why the switch appears disabled and the notice explains how to resolve it. A parent class marked as Multi-instance does not force its children to be so: each child keeps its own mode. The "0 wrng" counter at the top of the Inspector indicates the number of warnings for the selected element.

The available stereotypes are CoManagedEntity, SelfManagedEntity and Abstract. CoManagedEntity can produce an AAS Entity; SelfManagedEntity and Abstract are kept in UML but are not generated as main AAS elements. The Multi-instance option only appears on classes that take part in the AASX export.

4.5 Editing attributes — Attributes tab

The Attributes tab of the Inspector lists the attributes defined for the selected class, with their name and an icon to delete them. The counter next to the "Attributes" header indicates how many attributes the class has.

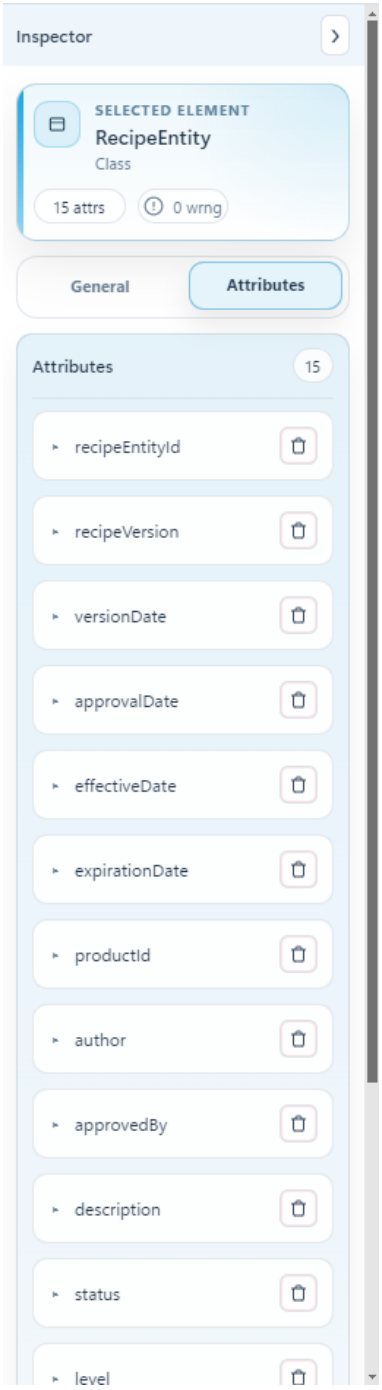


Figure 11. Inspector, Attributes tab, showing the 15 attributes of the RecipeEntity class.

To add a new attribute, use the corresponding control at the end of the attribute list; to delete one, click the trash icon next to its name. Each attribute can be expanded (arrow to the left of the name) to configure its data type and other properties.

- **Visibility:** select public (+), private (-), protected (#) or package (~).
- **Name:** must be a valid AAS idShort when the attribute is exported.
- **Type:** choose String, Integer, Boolean or Double (Primitive types), or a class/enumeration already defined in the project (Project types). The Project types list does not include the class itself, to avoid

accidental self-references. Custom... lets you type another type, but validation may block it if it does not exist in the project.

- **Multiplicity:** choose 0..1, 1, 0..* or 1..*, or Custom... to define your own range. In that case, enter Minimum and Maximum, or check Unlimited maximum (*) to leave the maximum open (for example, 2..*). The UML line on the card shows the result. If left empty, the export uses 1 and shows a warning.
- **Static and Abstract:** affect the UML notation of the attribute; on their own they do not create a different AAS type.

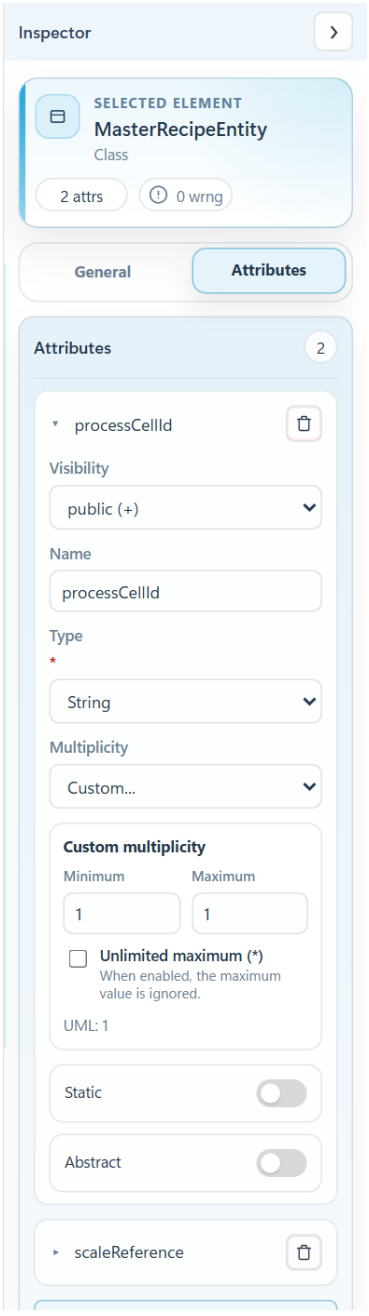


Figure 12. Expanded attribute in the Attributes tab with the custom multiplicity editor.

> **Note:** the custom multiplicity editor ignores invalid values (empty, negative or with a maximum lower than the minimum) and, when leaving the field, restores the last valid value. If the minimum is raised above the

> maximum, the maximum is adjusted automatically. In AASX, custom bounds are written in English in the cardinality qualifier: for example, 2..7 is exported as TwoToSeven and 2..* as TwoToMany.

4.6 Enumerations and notes

- When an Enumeration is selected, open the Literals tab and use Add Literal. Literals become AllowedValue when the enumeration is the type of a class attribute.
- When a Note is selected, edit its content in General > Note Text and link it with Note Link if you need to associate it with an element.

> **Note:** an Enumeration or a Note is not generated as a main AAS element; its role in AASX is auxiliary or purely for documentation.

4.7 Creating relations between classes

To link two classes, choose the relation type in the Relations panel and click first on the source class and then on the target class. Class Studio automatically validates the created relation; if it detects a problem (for example, a relation without a label), the line changes colour on the canvas and the issue appears in the Validation panel.

- Association: select the source first and then the target.
- Aggregation and Composition: select the whole/owner first (where the diamond is drawn) and then the part.
- Generalization: select the parent class first and then the child; the triangle is placed at the parent class end.
- Specialization: select the child class first and then the parent; the triangle is placed at the parent class end.

While you are drawing the relation, Escape cancels the connection. Once created, Swap direction (section 4.8) lets you invert source and target without redrawing it.

4.8 Editing relations and their routing

- Select a relation and use Relationship to change Type or Label, and Swap direction to swap Source and Target.

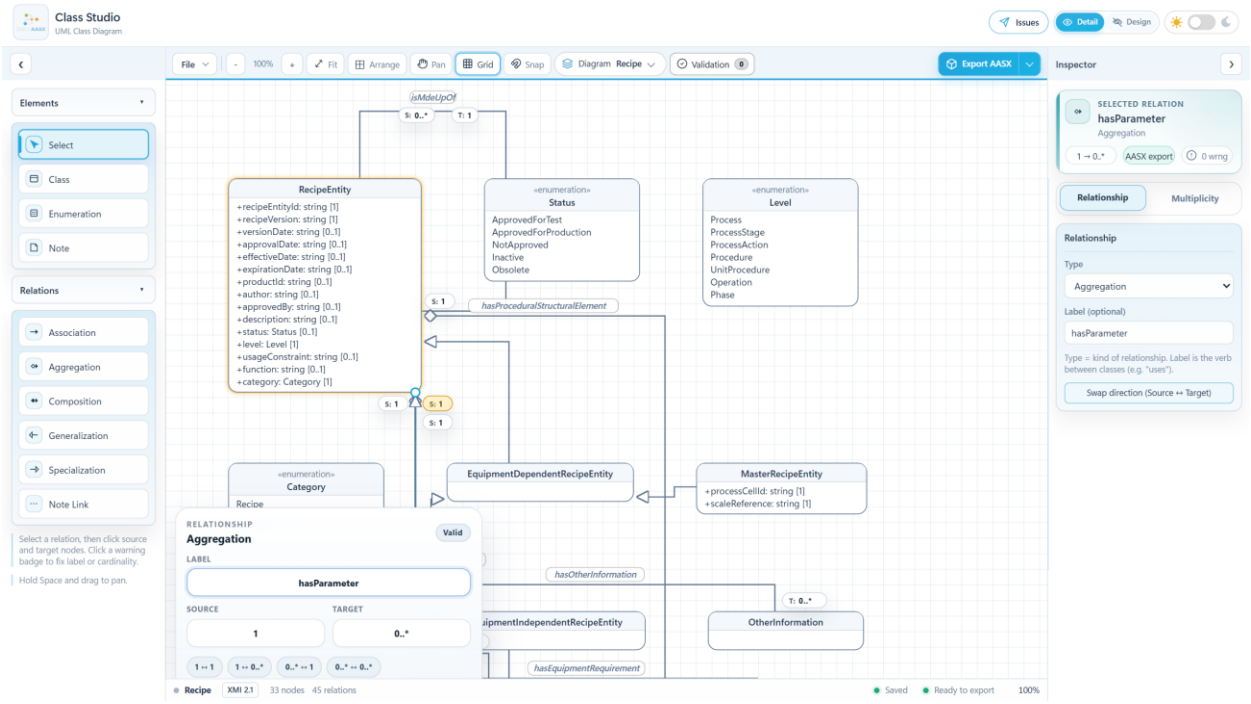


Figure 13. Selected hasParameter relation: floating panel on the canvas and Relationship tab of the Inspector.

- In Multiplicity, choose a pattern 1↔1, 1↔0..*, 0..*↔1 or 0..*↔0..*, or configure each end. Only Association, Aggregation and Composition use multiplicity and are exported as RelationshipElement.
- In Detail mode you can also edit the label and multiplicities in the relation's floating panel or by clicking their indicators.
- Manual route adjustment: drag a segment or a control point to modify the orthogonal path; if a segment would pass over another class, the line automatically goes around it. Drag an end and drop it on another node (or on one of its sides) to reconnect it on that side.
- Labels and multiplicity chips are placed automatically next to the line, without overlapping each other or the classes. They can also be dragged to another position.
- A manually adjusted relation keeps its shape when classes are moved and is saved with the project; routes imported from XMI or draw.io are preserved in the same way. To return to automatic routing, select the relation and click **Reset to automatic route** in the Relationship tab of the Inspector (Figure 14). The action can be undone.

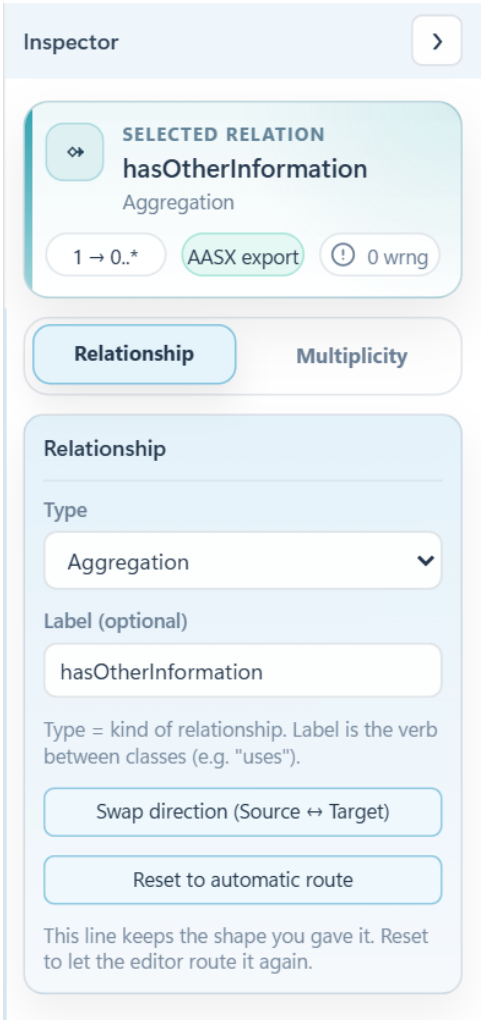


Figure 14. Relation with a manual route: Reset to automatic route button in the Inspector.

4.9 Keyboard and mouse shortcuts

- Ctrl+S: save the project as JSON.
- Ctrl+C / Ctrl+V: copy and paste the selection.
- Delete: delete the selected nodes or relation.
- Ctrl+Z / Ctrl+Y: undo and redo. Bulk actions, such as Arrange or Apply all in the validation panel, are undone in a single step.
- Ctrl+D: clear the current selection; it does not duplicate elements.
- Space + drag or middle button + drag: pan the canvas. Wheel: zoom in or out.
- Escape: cancel a connection or a direct edit in progress.
- During manual adjustment of a relation, the arrow keys move the control by 1 pixel; with Shift, 5 pixels.

5. Saving, opening and importing projects

All workspace, import, save and export operations are grouped in the File menu, in the top left corner of the toolbar.

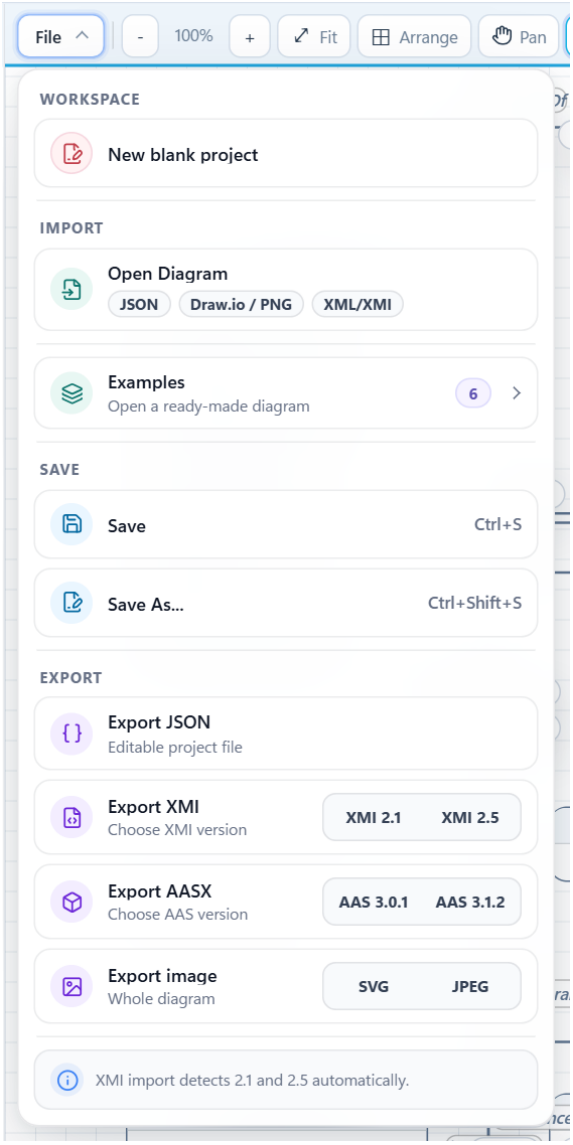


Figure 15. File menu, with the Workspace, Import, Save and Export sections.

5.1 Workspace

- **New blank project:** discards the current diagram and opens a new empty one.

If there are unsaved changes, the tool asks for confirmation before discarding them. Validation issues do not prevent creating a new project: they only block the export to AASX.

5.2 Import

- **Open Diagram:** imports a JSON diagram, a .drawio file, a PNG that retains diagrams.net data, or an XML/XMI file.
- **Examples:** shows, within the same menu, the list of example diagrams (6 available, described in section 2). The top arrow (Back) returns to the main menu.

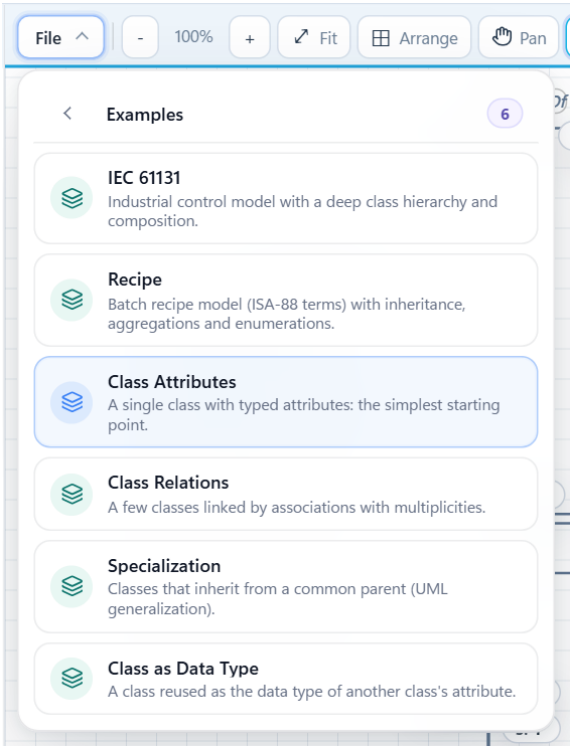


Figure 16. Examples view of the File menu.

> **Note:** when importing an XML/XMI file, the tool automatically detects whether it is XMI 2.1 or 2.5; there is no need to specify it manually.

When the import finishes, a notification confirms the imported file (and the detected XMI version) and the status bar shows the source format. If the diagram is large, it opens at 100% from the top left corner so the text is readable; use Fit to see it in full. Relation routes saved in the file are preserved.

5.3 Save

- **Save (Ctrl+S):** saves the current project.
- **Save As...:** saves the project with a new name or location. It suggests the diagram name followed by the date and time (for example, Recipe-20261026-101500.json).

> **Note:** the menu shows Ctrl+Shift+S next to Save As..., but this combination currently runs Save. To save with a different name, use File > Save As...

In the desktop application, Save and Save As open the operating system's save dialog for .json files. In the browser, the compatible picker is used or, if it is not available, the file is downloaded.

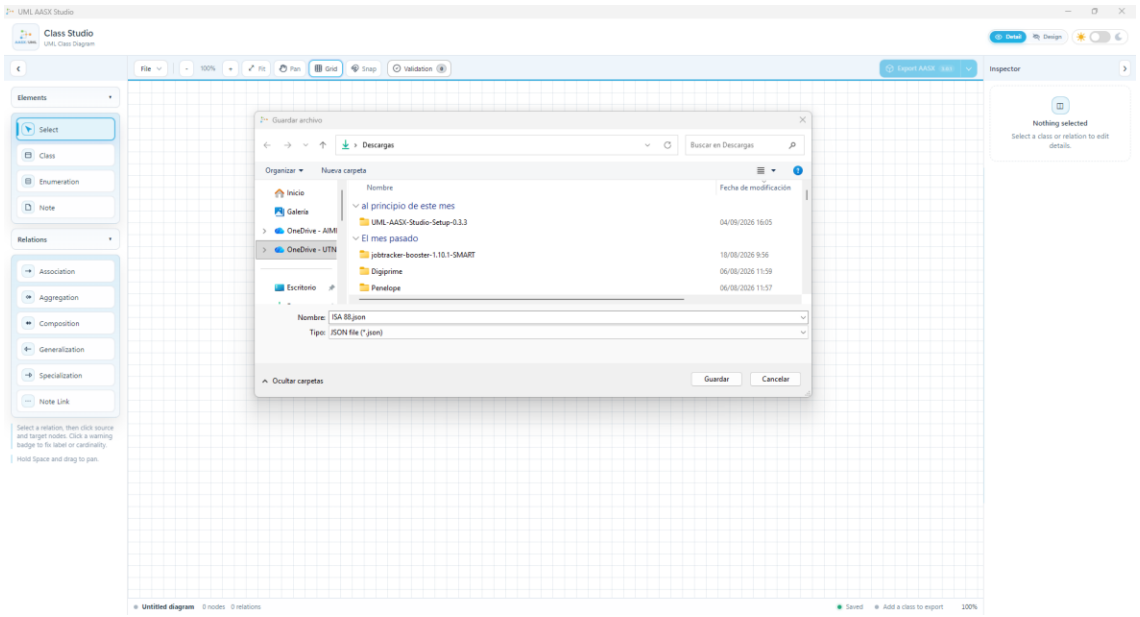


Figure 17. Operating system save dialog when using Save As.

> **Note:** the status bar changes to Unsaved changes after editing the model and returns to Saved when saving completes successfully. When opening another diagram or creating a new one with pending changes, the tool asks for confirmation before discarding them.

5.4 Export

Export JSON, Export XMI, Export AASX and Export image are available from the same File menu. Export JSON produces the same native model used by Save; XMI and AASX go through the backend and require it to be available.

- **Export JSON:** exports the model in the tool's native (editable) format.
- **Export XMI:** exports the model in XMI format, with a choice of XMI 2.1 or XMI 2.5.
- **Export AASX:** exports the model with the chosen AAS version (AAS 3.0.1 or AAS 3.1.2). It is equivalent to the Export AASX button on the toolbar (see section 7).
- **Export image:** generates an image of the full diagram, even if part of it is outside the view, in SVG (vector) or JPEG (high resolution) format. The image includes classes, relations, labels and multiplicities, does not show the selection or editing aids, and uses the colours of the active theme (light or dark).

In the desktop application, exported files are saved in the Downloads folder without overwriting existing files, and the final notification offers Open file to open the file directly. In the browser, the download is handled by the browser itself.

6. Model validation

Before exporting to AASX, Class Studio validates the UML model against the UML2AAS transformation rules. The Validation button on the toolbar shows the number of detected issues at all times. For this chapter, the Recipe example was used as a starting point, the labels of two relations were removed and a class was renamed "Master Recipe" (with a space, which is not valid in AAS).

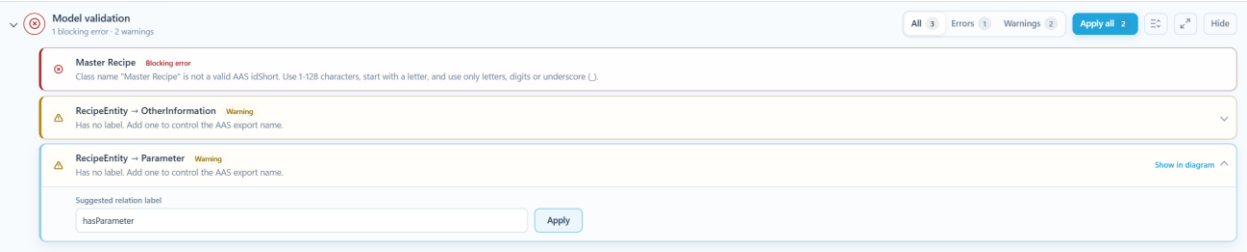


Figure 18. Model validation panel with one blocking error and two warnings; the expanded card suggests a label for the relation.

6.1 Issue types

Validation combines immediate checks of the UML model with a check of the AAS that the backend would generate. It runs again automatically after every change, so opening the panel does not alter the count. The counter always shows the total number of issues; while the backend check is in progress it may show an ellipsis.

Type	Meaning	Does it block the AASX export?
Warning	A situation that the tool can resolve with a default value (for example, a relation without a label, for which a fallback relation id will be used).	No
Blocking error	A situation that prevents generating a valid AASX (for example, an AAS relation name that exceeds the 64 characters allowed).	Yes

Common cases worth resolving:

- Errors: missing or unknown attribute type, invalid idShort, missing instance mode, hierarchy conflict, duplicate AAS names or a generated displayName longer than 64 characters.
- Warnings: missing relation label, missing or invalid multiplicity, or a UML relation that the current rules do not export as RelationshipElement. A Dependency towards a class that is already used as the type of an attribute of the source class does not generate a warning, because that attribute already represents the relation in AASX.
- The most common default values are multiplicity 1 and a relation label generated from the source, the type and the target.

6.2 How to fix issues

1. Open the Validation panel with the toolbar button or by clicking the AASX status in the status bar.
2. Review the list of issues; you can filter them by All, Errors or Warnings. Cards appear collapsed: click one to see its details.
3. Clicking an issue (or Show in diagram) selects the affected node or relation. The view only scrolls as much as needed to show them.
4. For issues about relations without a label, the card suggests a "Suggested relation label" (for example, hasParameter or hasOtherInformation). Click Apply to accept it, or type your own label before applying.
5. Use Apply all to accept all available suggestions at once; hovering over the button shows the labels that will be applied. The whole operation is undone in a single step with Ctrl+Z.

6. The header buttons let you expand or collapse all cards, maximize or restore the panel, and hide it (Hide).

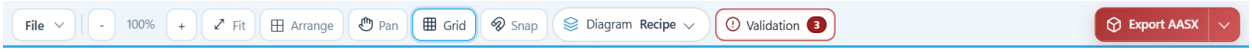


Figure 19. Toolbar with blocking errors: Validation and Export AASX are shown in red.

> **Caution:** while there are blocking errors, the Export AASX button is shown in red and, when clicked, opens the validation panel instead of exporting. Before generating the file, the backend checks again the AAS names that will be created (for example, displayName longer than 64 characters or duplicate idShort). Warnings do not block the export, but it is advisable to review them because they determine the names the AAS Submodel will finally have.

6.3 Collapsed panel: issue navigator

When the panel is collapsed with the arrow on the left of its header, a bar remains that shows the active issue, its severity and its position (for example, Warning 2 / 3). The < and > arrows cycle through the issues of the current filter and select each element in the diagram without reopening the panel. If the active issue is fixed, the bar moves to the first visible issue.



Figure 20. Collapsed validation panel, showing the second of three issues.

When the panel is maximized, the card of the active issue is expanded and kept in view.

7. Exporting to AASX

Once the model has no blocking errors, the AASX package (.aasx) corresponding to the AAS Submodel can be generated.

7.1 Choosing the AAS version

Expanding the arrow next to the Export AASX button lets you choose the AAS metamodel version to use:

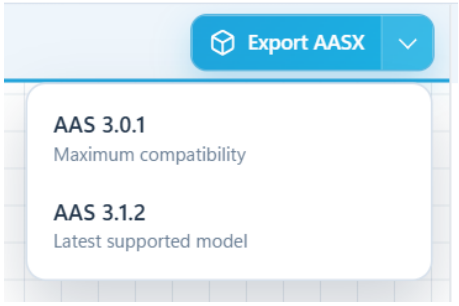


Figure 21. AAS version selection when exporting.

- **AAS 3.0.1 — Maximum compatibility:** recommended when the AASX must be opened with external tools that do not yet support the latest version.
- **AAS 3.1.2 — Latest supported model:** latest version supported by the tool; recommended by default.

7.2 Naming and confirming the export

When the export starts, a name is requested before querying the backend. Its base is used for the .aasx file and for the Submodel idShort; the AAS idShort is derived by adding the suffix AAS (for example, Recipe and RecipeAAS).

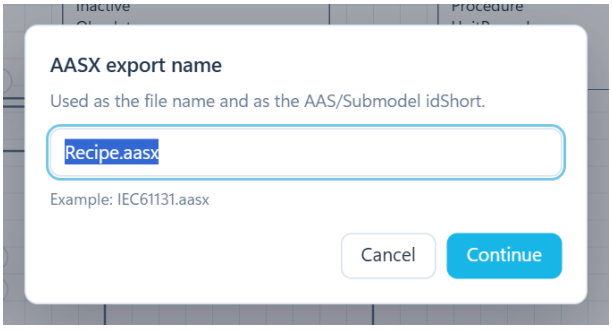


Figure 22. AASX export name dialog.

1. Type the desired name (for example, Recipe.aasx or IEC61131.aasx).
2. Click Continue to generate and download the file, or Cancel to return to the canvas without exporting.

> **Note:** choose a short name that starts with a letter and uses only letters, numbers or underscore (_). If the value is not valid, the dialog stays open and suggests a corrected version.

7.3 What is actually transformed to AASX

The editor can keep more UML information than the exporter transforms. The current scope is as follows:

- A Class is transformed into a SubmodelElementCollection; if Multi-instance is enabled, it is represented by a SubmodelElementList.
- A Class with CoManagedEntity can be transformed into an Entity when it meets the eligibility rules. SelfManagedEntity and Abstract classes are not generated as main AAS elements.
- Primitive attributes are converted into Properties. An attribute typed with an Enumeration receives AllowedValue qualifiers with its literals.
- Multiplicities are converted into cardinality qualifiers (One, ZeroToOne, ZeroToMany, OneToMany). Custom ranges are written with the bounds in English, for example TwoToSeven for 2..7 or TwoToMany for 2..*.
- Generalization and Specialization are used to inherit attributes and organize containers; Association, Aggregation and Composition generate AAS relations. Composition can also produce structural nesting.
- Each exported relation generates a list of relations whose name combines source, label and target. If several relations repeat the same source, label and target, their names are numbered (for example, uses, uses2) so that they are unique; that is why it is advisable to label all relations.
- Note and Note Link remain as UML documentation. Other imported types that are not within the scope of the rules are kept in the model but are not converted into main AAS elements.

7.4 Result and file location

The generated package contains the AAS environment, the Submodel and auxiliary resources of the diagram. In the desktop application it is saved in Downloads without silently overwriting an existing file, and the final notification offers Open file. In the web version, the browser handles the download.

8. Recommended workflow

1. Start a new project or import an existing diagram (sections 2 and 5.2).
2. Model the domain's classes, enumerations, attributes and relations (section 4).
3. Define the Single/Multi-instance mode of each exportable class, the type and multiplicity of each attribute, and the direction, label and cardinalities of the AASX relations.
4. Tidy up the diagram with Arrange or by adjusting relations by hand, if needed for better readability (sections 3.3 and 4.8).
5. Save your progress periodically with Save / Save As (section 5.3).
6. Open the Validation panel and resolve warnings and blocking errors (section 6).
7. Choose the appropriate AAS version and export the AASX (section 7).
8. Verify the resulting AASX with an external tool (for example, AASX Package Explorer or the AIMEN toolkit) to confirm its compliance.
9. Optionally, export an SVG or JPEG image of the diagram for the project documentation (section 5.4).

9. Reporting problems and suggestions

The **Issues** button in the top bar opens the **Report & Feedback**form, which prepares an email to the UML AASX Studio team with a bug report or an improvement proposal.

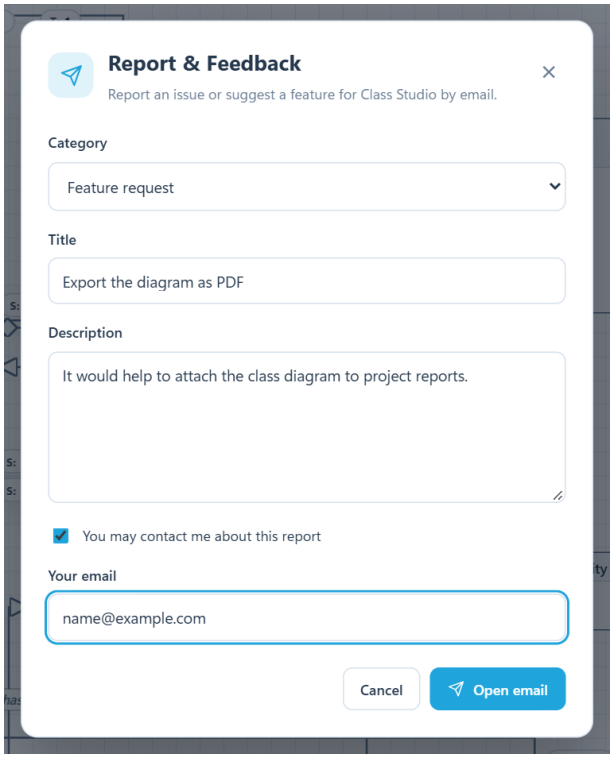


Figure 23. Report & Feedback form.

- **Category:** Issue (problem) or Feature request (improvement proposal).

- **Title:** short summary (up to 100 characters).
- **Description:** what happened or what you would like to improve (up to 1500 characters). For a problem, give the steps to reproduce it and the diagram or example used.
- **You may contact me about this report:** if you check it, enter your email in Your email so the team can reply to you. It is only included in the message if you choose so.

Clicking **Open email**, the tool opens your email client with the recipient (umltoaasxstudio@gmail.com), subject and body already filled in. The message is not sent automatically: review it and send it from your email client. If no client opens, you can write directly to that address.

10. Glossary

Term	Description
AAS	Asset Administration Shell: standardized digital representation of an asset.
Submodel	AAS structure that groups a set of properties and data related to one aspect of the asset.
SMC	Submodel Element Collection: collection of submodel elements, used to represent a UML class.
AASX	Package format (compressed file) that contains one or more AAS and their submodels.
idShort	Short, readable identifier of an AAS element; it is derived from the name of the exported file.
Multi-instance	Property of a class indicating that it must be exported as a list of SMCs instead of a single SMC.
XMI	XML Metadata Interchange: standard format for exchanging UML models.
Blocking error	Validation issue that prevents generating a valid AASX until it is fixed.
SubmodelElementList (SMC List)	AAS list used to represent elements or classes with multiple instances.
CoManagedEntity	UML stereotype that allows an eligible class to be transformed into a co-managed AAS Entity.
AllowedValue	Qualifier generated from the literals of an Enumeration when it is used as the type of an attribute.
Generalization / Specialization	Two visual orientations of inheritance supported by the editor; they are used to resolve inherited attributes and AASX containers.
Arrange	Toolbar action that automatically reorganizes the classes in layers according to their inheritance, composition and aggregation relations.
Manual route	Path of a relation adjusted by the user or imported; it is kept when classes are moved until Reset to automatic route is clicked.
Cardinality (SMT/Cardinality)	AAS qualifier generated from the UML multiplicity (for example, One, ZeroToMany or TwoToSeven).

Term	Description
Design mode	Canvas presentation view that hides the technical aids without modifying the model.